

Chapter 5:

Newton's Laws of Motion

From **kinematics**, the description of motion,
to **dynamics**, the study of causes of changes
in motion.

Newton's Laws of Motion



Helpful Videos

- Newton's Laws – Physics Concept Trailer:
<https://www.youtube.com/watch?v=lCxxH8nQtZQ>
- Dynamics: What are Forces?
<https://www.youtube.com/watch?v=iAMxWvyykRU>
- Newton's First Law of Motion: Mass and Inertia
<https://www.youtube.com/watch?v=1XSyyjcEHo0>
- Newton's Second Law of Motion: $F = ma$
<https://www.youtube.com/watch?v=xzA6IBWUEDE>
- Newton's Third Law of Motion: Action and Reaction
https://www.youtube.com/watch?v=y61_VPKH2B4&vl=sr
- Mass vs. Weight and the Normal Force
<https://www.youtube.com/watch?v=57gKnA94Doc>
- Frictional Forces: Static and Kinetic
<https://www.youtube.com/watch?v=3EbUa5ZDybg>

Chapter 5: Outline

5.1 Force and Mass

Concepts

5.2 Newton's First Law

5.3 Newton's Second Law

5.4 Newton's Third Law

Physics Laws
& Formulas

5.5 Common Forces

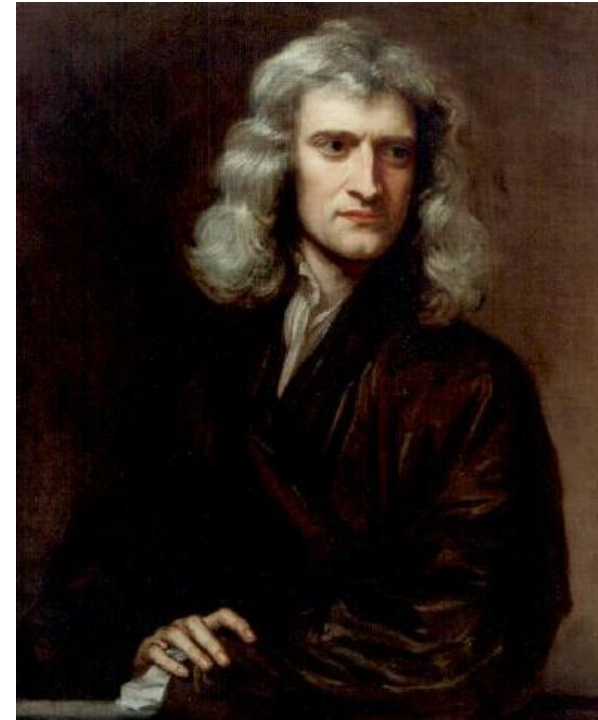
5.6 Drawing Free-Body Diagrams

5.7 Solving Problems with Newton's Laws

Problem Solving

Kinematics vs Dynamics

- **Kinematics** only describes the way objects move — their position, velocity, and acceleration (Chapter 3-4).
- **Dynamics** is the study of how forces affect the motion of objects and systems.
- The foundation of dynamics are the laws of motion stated by Isaac Newton (1642–1727).

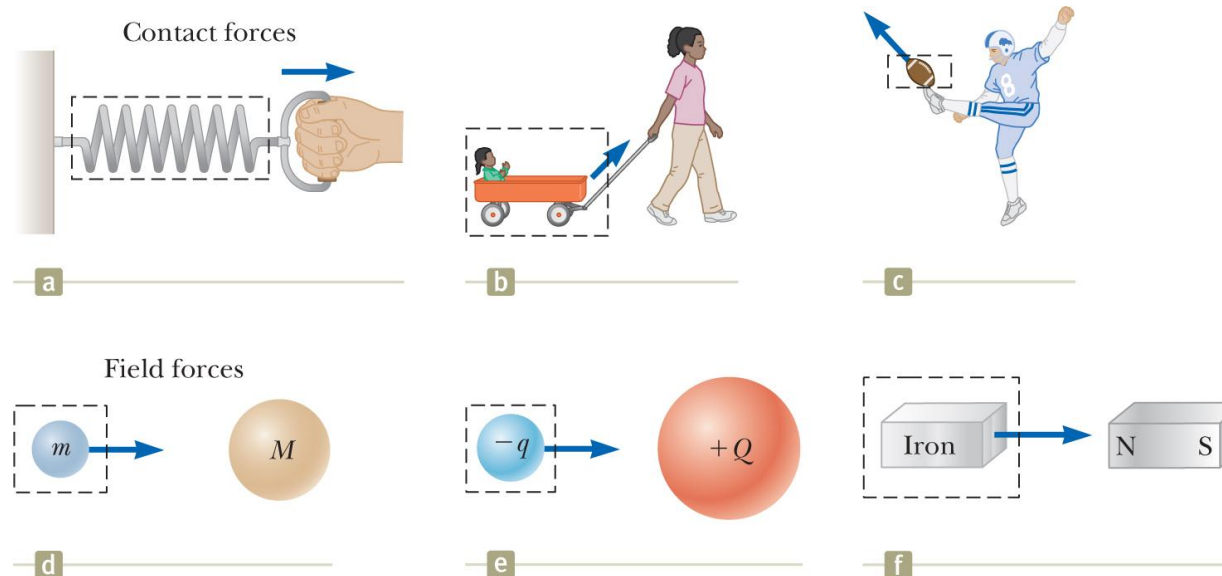


5.1 Force and Mass

- What is force? Scalar or vector?
- Does force always involve physical contact?
- How do you draw forces acting on an object?
- What is mass? Scalar or vector?
- Are mass and weight the same?

Fundamental Forces

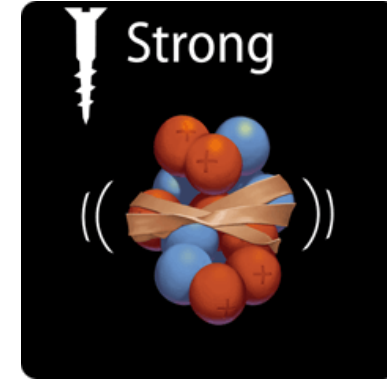
- **Force** refers to interaction with object to change its velocity. We can define force as the *push or pull on an object* with a specific magnitude and direction.
- **Contact Forces:** involve physical contact between two objects.
- **Field Forces:** no physical contact between two objects → act through empty space (such as electric and magnetic fields).



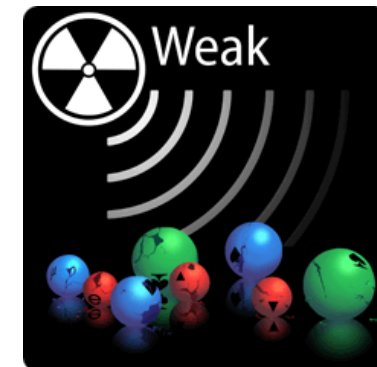
Fundamental Forces

- **Fundamental forces in nature:**

1. *Gravitational forces* between objects
2. *Electromagnetic forces* between electric charges
3. *Strong forces* between subatomic particles
4. *Weak forces* that arise in certain radioactive decay processes

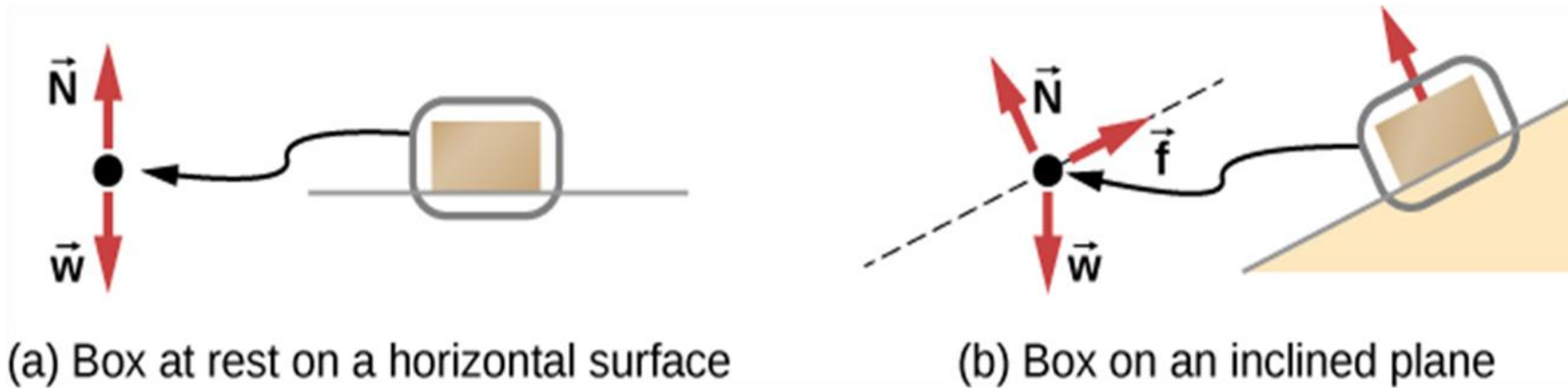


- Classical physics → only concerned with gravitational and electromagnetic forces



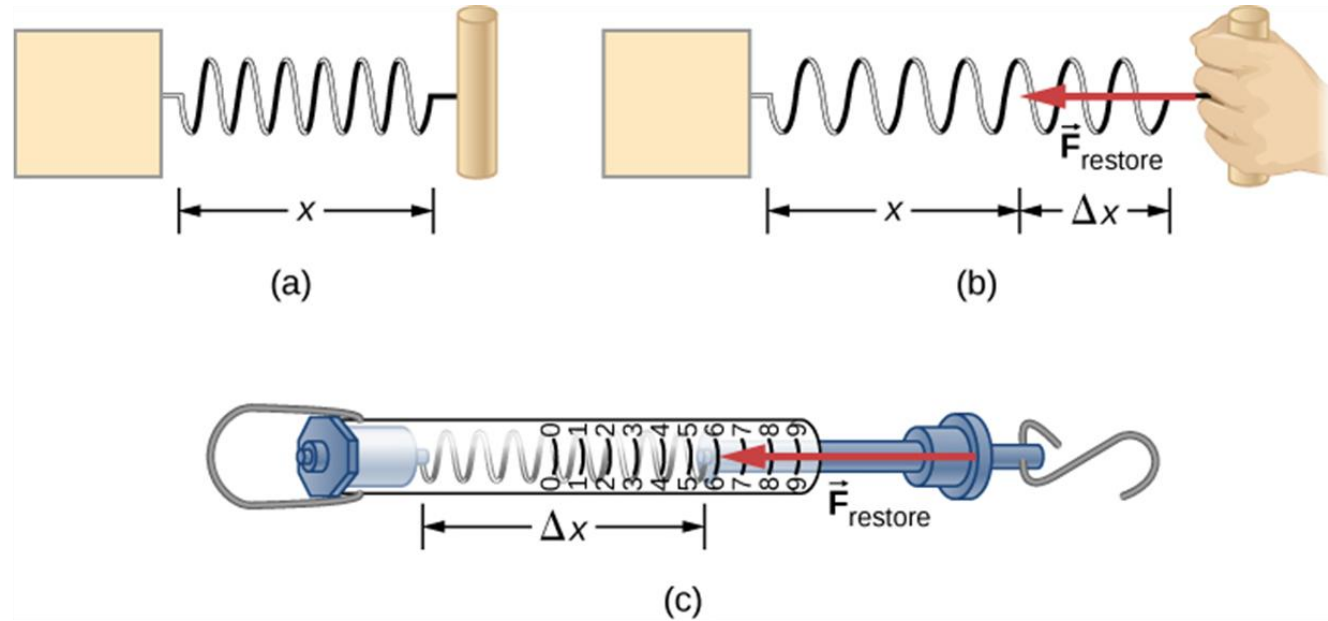
Free Body Diagram (FBD)

- Free-body diagram: a sketch showing all external forces acting on an object or system.
- Free-body diagrams are useful in analyzing forces acting on an object or system and are employed extensively in the study and application of Newton's laws of motion.



In these free-body diagrams, $\vec{N}$ is the normal force, $\vec{w}$ is the weight of the object, and $\vec{f}$ is the friction.

How to measure Force



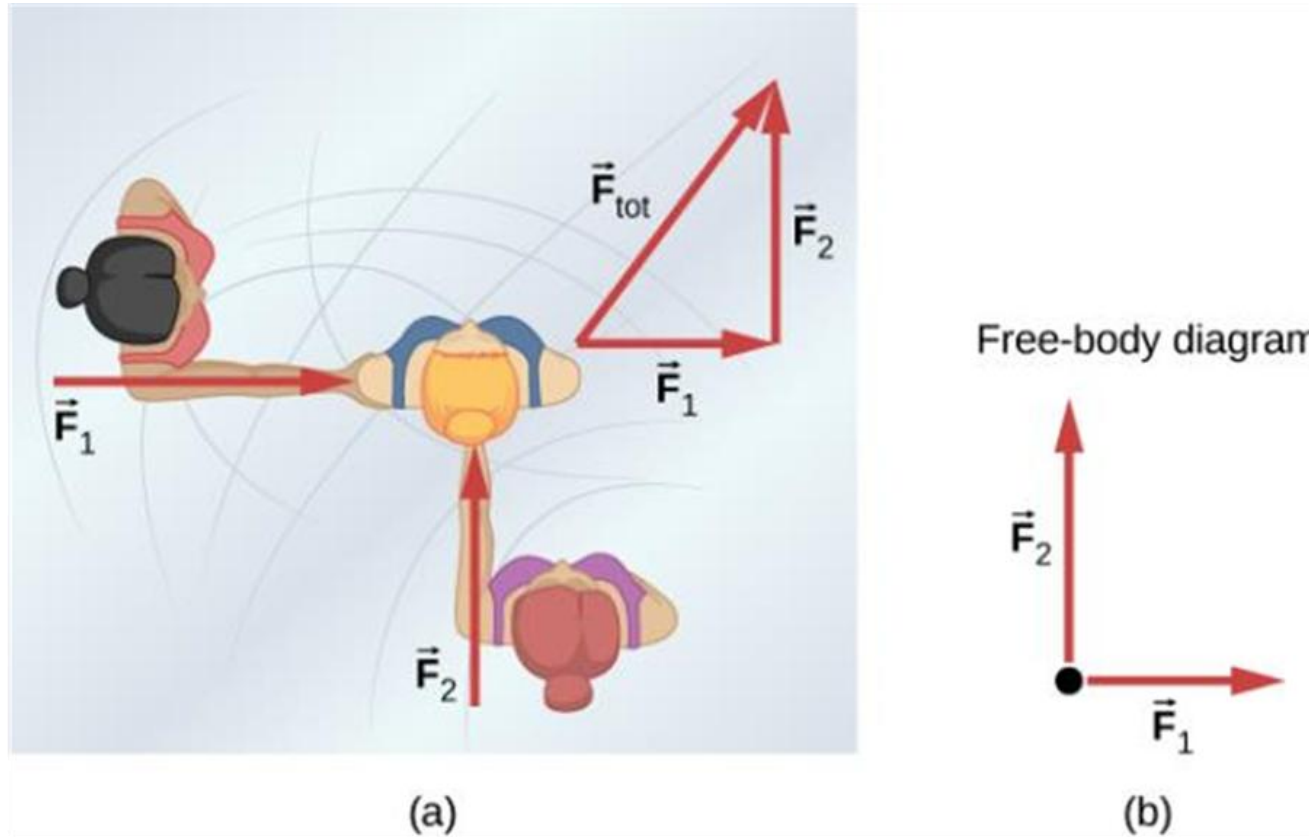
The force exerted by a stretched spring can be used as a standard unit of force.

(a) This spring has a length x when undistorted.

(b) When stretched a distance Δx , the spring exerts a restoring force $\vec{F}_{\text{restore}}$, which is reproducible.

(c) A spring scale is one device that uses a spring to measure force. The force $\vec{F}_{\text{restore}}$ is exerted on whatever is attached to the hook.

Forces as Vectors



(a) An overhead view of two ice skaters pushing on a third skater. Forces are vectors and add like other vectors, so the total force on the third skater is in the direction shown.

(b) A **free-body diagram** representing the forces acting on the third skater.

Force: A Vector Quantity

- **Must** use rules of vector addition to obtain the net force on an object.

$$|\vec{F}| = \sqrt{F_1^2 + F_2^2}$$

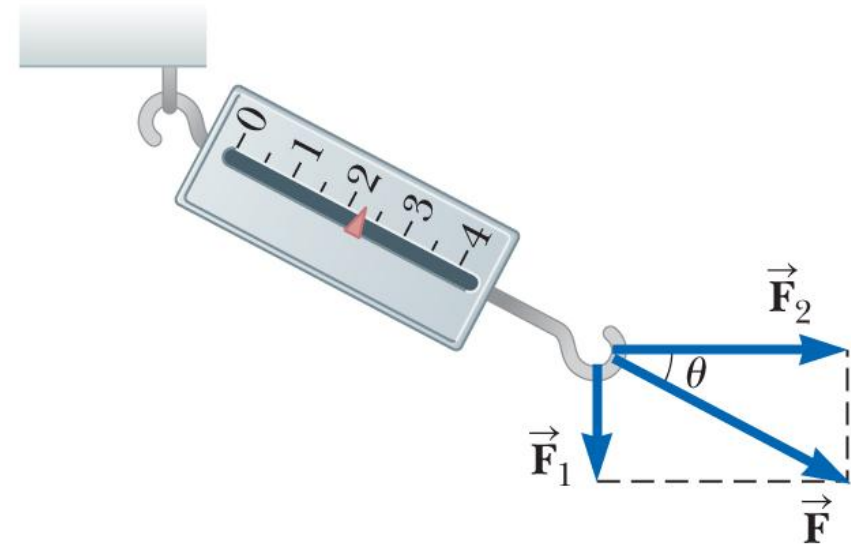
$$= 2.24 \text{ units}$$

$$\theta = \tan^{-1} \left(\frac{1.00 \text{ unit}}{2.00 \text{ unit}} \right)$$

$$= \tan^{-1}(0.500)$$

$$= 26.6^\circ$$

When $\vec{F}_1$ is downward and $\vec{F}_2$ is horizontal, the combination of the two forces elongates the spring by 2.24 cm.



Units of Forces

SI unit of force: **newton** (N)

$$1 \text{ N} \equiv 1 \text{ kg} \cdot \text{m/s}^2$$

$$1 \text{ lb} \equiv 1 \text{ slug} \cdot \text{ft/s}^2$$

$$1 \text{ N} \approx \frac{1}{4} \text{ lb}$$

Mass

- **Mass** is a property that specifies how much resistance an object exhibits to changes in its velocity.
- Experiments show: greater the mass of an object → less that object accelerates under action of given applied force.
- Magnitude of acceleration of object inversely proportional to its mass when acted on by *a given force*.

$$\frac{m_1}{m_2} \equiv \frac{a_2}{a_1}$$

- Mass is a scalar quantity (SI unit: kg) and inherent property of an object, independent of the object's surroundings and of method used to measure it.

Inertial Frame of Reference

If an object does not interact with other objects, it is possible to identify a reference frame in which the object has zero acceleration.

Inertial frame of reference

- We model Earth as inertial frame, along with any other frame attached to it in our course.
- Video 1: <https://www.youtube.com/watch?v=wD7C4V9smG4>
- Video 2: <https://www.youtube.com/watch?v=8xiIP2B6L3M>

5.2 Newton's **First** Law

Group Discussion:

- What are the Newton's Laws of Motion?
- What is the equation for Newton's Second Law?
What are the quantities in the equation? Is it a vector or scalar equation?
- Are Newton's Laws of Motion consistent with or contradicting our daily life experience?

5.2 Newton's **First** Law

A body at rest **remains** at rest or, if in motion, remains in motion at constant velocity unless acted on by a net external force.

- When no net force acts on an object, the acceleration of the object is zero (rest or constant velocity).

$$\vec{v} = \text{constant when } \vec{F}_{\text{net}} = \vec{0} \text{ N.}$$

- **Inertia:** the tendency of an object to resist any attempt to change its velocity.

If $\vec{F}_{net} = 0$ (no net force acting on the body)

- and the body is at rest



then the body remains with $v = 0$.

- and the body is moving with constant velocity



then the body remains moving with constant velocity.



Even today people still believe that a force is required to keep an object moving.

Objects keep on doing what they're doing.



Accidents



Inertia is tendency of an object to change its velocity.

It is related to mass.

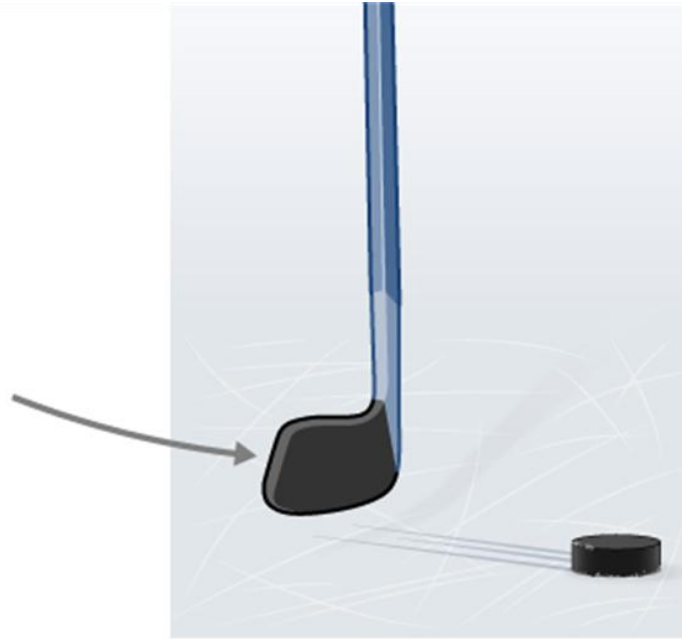
The bigger the mass, more resistance to changes in the motion.



NOT contradicting our experience



(a)



(b)

**Friction or
resistance
everywhere!**

- (a) A hockey puck is shown at rest; it remains at rest until an outside force such as a hockey stick changes its state of rest;
- (b) A hockey puck is shown in motion; it continues in motion in a straight line until an outside force causes it to change its state of motion. Although it is slick, an ice surface provides **some friction** that slows the puck.

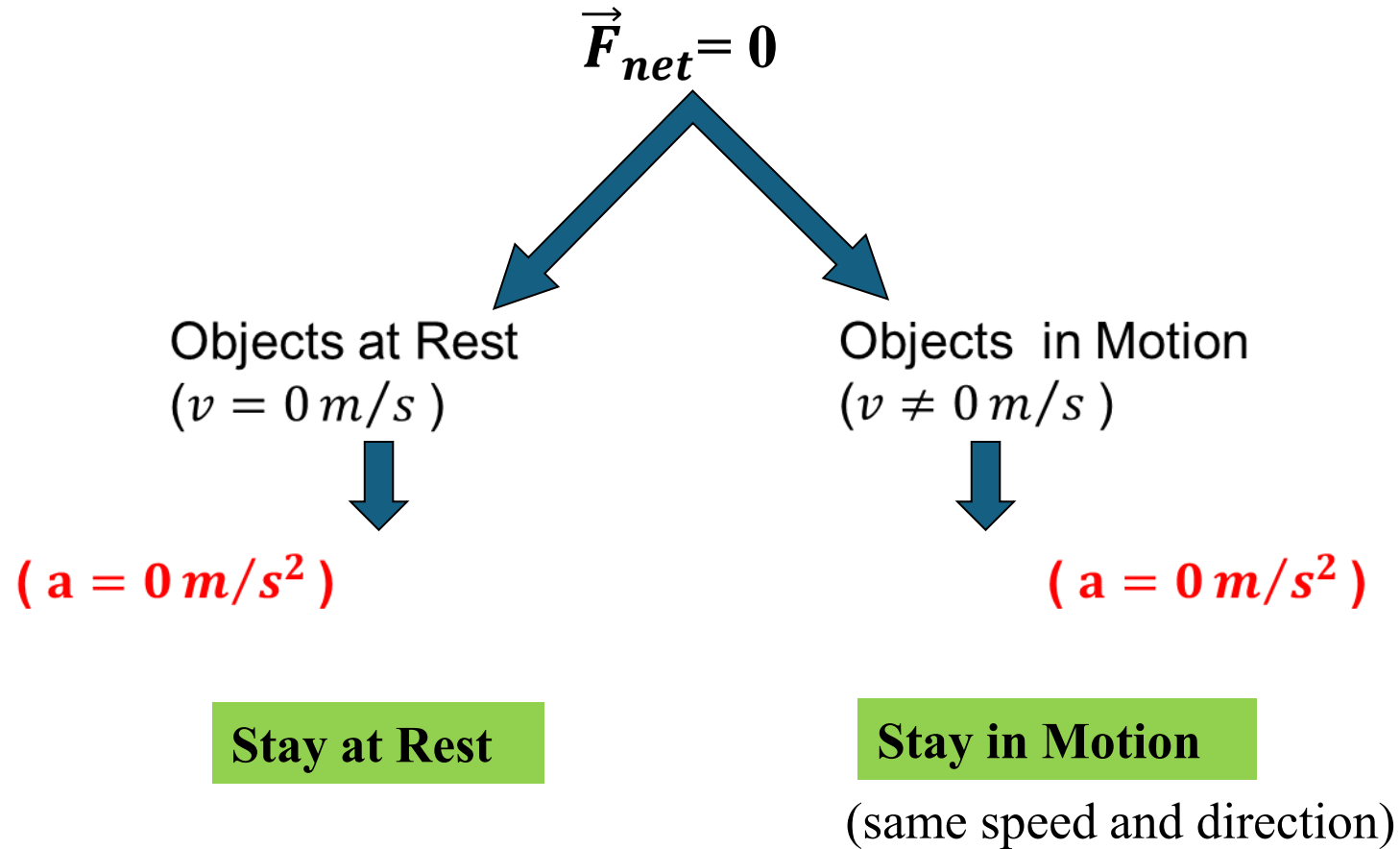
Newton's **First** Law: Hockey Puck



Newton's **First** Law



Newton's First Law



What does " $\vec{F}_{net} = 0$ " mean??

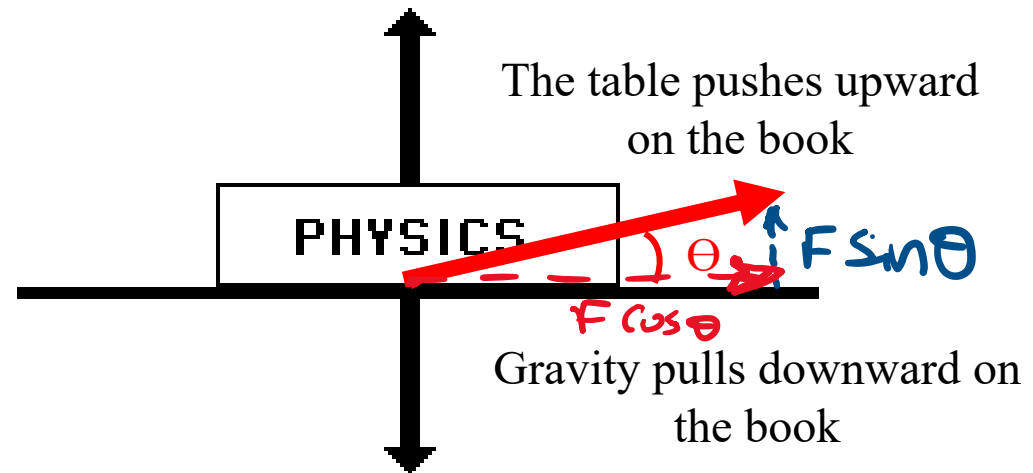
acceleration = 0 if $\vec{F}_{net} = 0$

Newton's First Law

The forces on the person are balanced.



The forces on the book are balanced.



What if a Force (F) is applied at an angle?

$$\vec{N} + F \sin \theta = \vec{W} (mg)$$

$$\vec{N} = \vec{W} - F \sin \theta$$



Two equal-magnitude horizontal forces act on a box. Is the object **accelerated** horizontally?

- a) Yes
- b) No
- c) You can't tell from this diagram.

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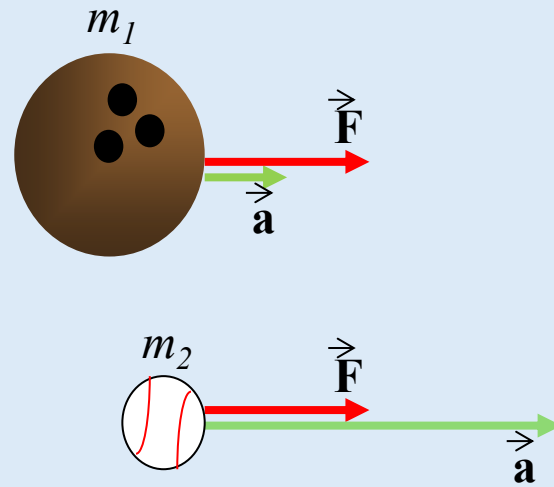


Is it possible that the box is **moving**, when two forces equal in size but opposite in direction are applied?

- a) Yes, it is possible for the object to be moving.
- b) No, it is impossible for the object to be moving.



5.3 Newton's **Second** Law



The smaller-mass object experiences a larger acceleration than the larger-mass object when identical forces are applied to the two objects.

5.3 Newton's Second Law

Newton's Second Law of Motion

The acceleration of a system is directly proportional to and in the same direction as the net external force acting on the system and is inversely proportion to its mass. In equation form, Newton's second law is

$$\vec{\mathbf{a}} = \frac{\vec{\mathbf{F}}_{\text{net}}}{m},$$

where $\vec{\mathbf{a}}$ is the acceleration, $\vec{\mathbf{F}}_{\text{net}}$ is the net force, and m is the mass. This is often written in the more familiar form

$$\vec{\mathbf{F}}_{\text{net}} = \sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}, \quad 5.3$$

but the first equation gives more insight into what Newton's second law means. When only the magnitude of force and acceleration are considered, this equation can be written in the simpler scalar form:

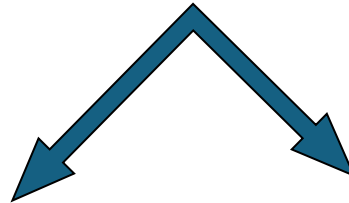
$$F_{\text{net}} = ma. \quad 5.4$$

5.3 Newton's **Second Law**

Net-Force non-zero



There is an acceleration



The acceleration depends
directly upon the
“net force”

The acceleration depends
inversely upon the
“object’s mass”

What does “ $\vec{F}_{net} \neq 0$ ” mean??

$$\vec{a} \neq 0$$

Component Form of Newton's **Second Law**

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

- This vector equation can be written as three component equations:

$$\sum F_x = ma_x \quad \sum F_y = ma_y \quad \sum F_z = ma_z$$

- Use Component Form equations when solving problems.

Newton's **Second** Law: Hockey Puck



Newton's Second Law

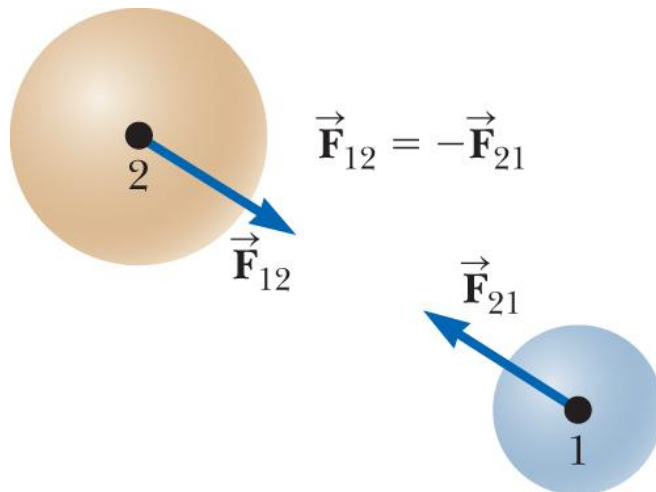


5.4 Newton's Third Law

Newton's Third Law of Motion

Whenever one body exerts a force on a second body, the first body experiences a force that is equal in magnitude and opposite in direction to the force that it exerts. Mathematically, if a body A exerts a force $\vec{F}$ on body B , then B simultaneously exerts a force $-\vec{F}$ on A , or in vector equation form,

$$\vec{F}_{AB} = -\vec{F}_{BA}. \quad 5.10$$



- Every force has an agent and acts on an object.
- Identify the **agent** and **object** for each force.

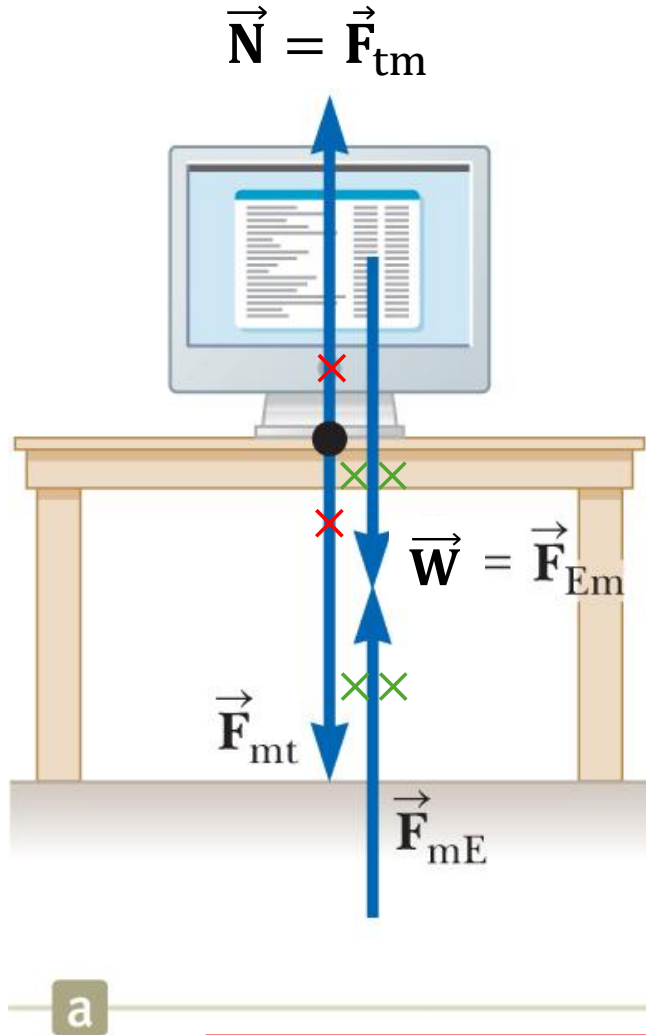
Newton's **Third** Law: Hockey Puck



Newton's Third Law



Newton's Third Law



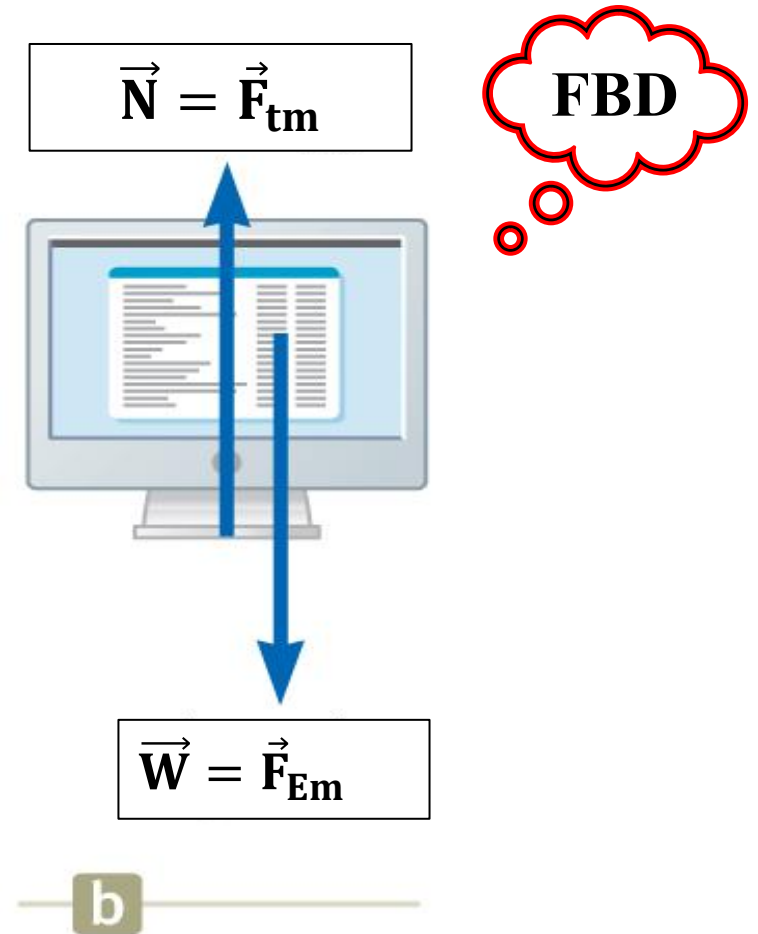
Monitor is at rest,

$$\sum \vec{F} = 0$$

$$\vec{N} + m\vec{g} = 0$$

$$\Rightarrow N\hat{j} - mg\hat{j} = 0$$

$$N = mg$$



$\vec{F}_{tm}$ = force by table **ON** monitor

$\vec{F}_{Em}$ = force by Earth **ON** monitor

$\vec{F}_{mt}$ = force by monitor **ON** table

$\vec{F}_{mE}$ = force by monitor **ON** Earth

Gravitational Force ($\vec{F}_g$) and Weight ($\vec{W}$)

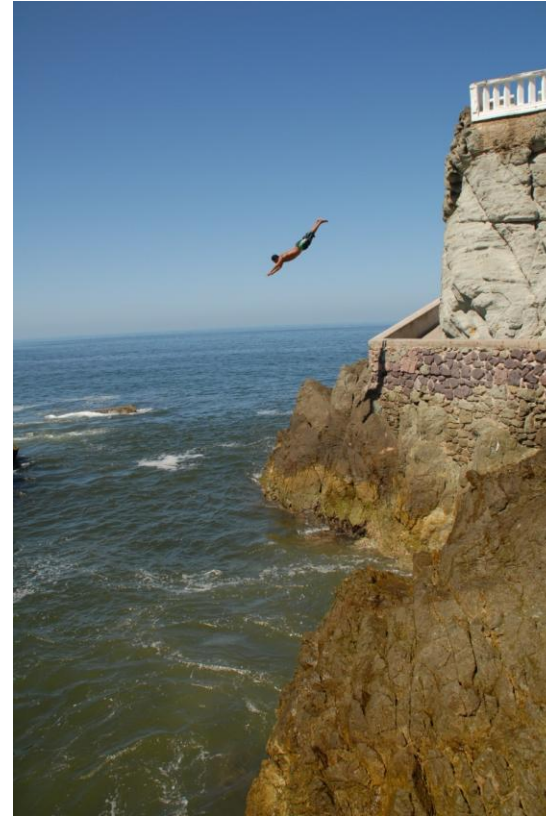
- **Gravitational force:** attractive force exerted by Earth on object
- Directed toward center of Earth

$$\vec{F}_g \text{ or } \vec{W} = m \vec{g}$$

- $g \sim 9.80 \text{ m/s}^2$
- g varies with geographic location;
- g decreases with increasing distance from center of Earth.

$$F_g = mg_{\text{Moon}}$$

$$g_{\text{Moon}} \approx 1.6 \text{ m/s}^2$$



Normal Force ($\vec{N}$)

Definition: the force that surfaces exert to prevent solid objects from passing through each other

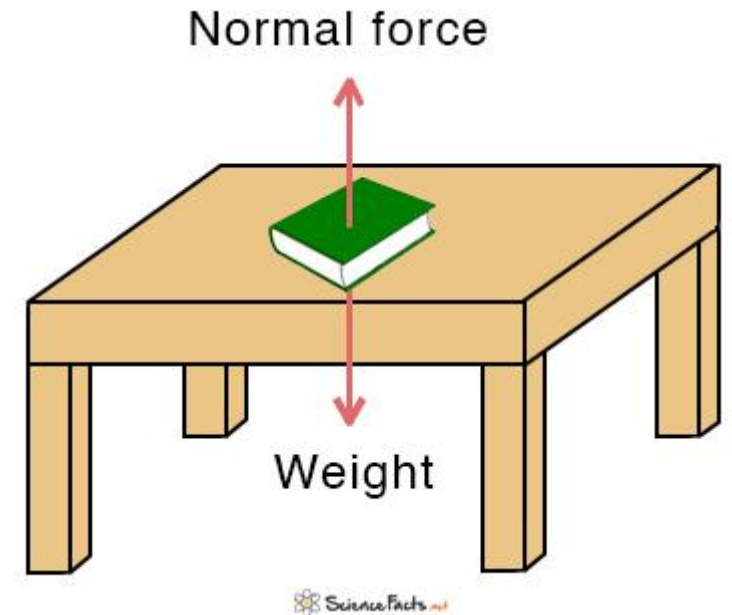
Variable: $\vec{N}$

Unit: Newtons (N)

Equation: No specific equation

- $N = F_g$ if $a_y = 0 \text{ m/s}^2$
- $N \neq F_g$ if $a_y \neq 0 \text{ m/s}^2$

Direction: Pointed away from surface & perpendicular to the surface



When is $\vec{N} \neq \vec{F}_g$ or $\vec{W}$

$$\sum \vec{F}_y = m\vec{a}_y$$

$$\sum \vec{F}_y = 0 \text{ (rest or equilibrium)}$$

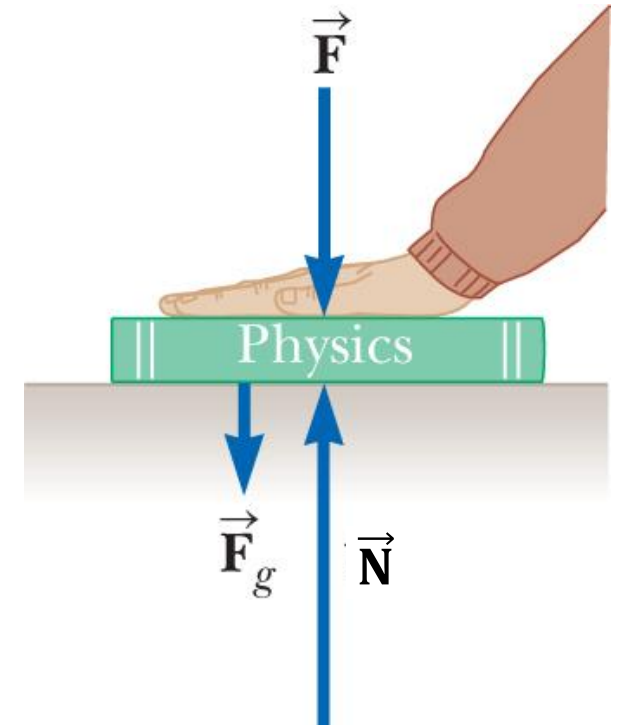
$$N \hat{j} - F \hat{j} - F_g \hat{j} = 0$$

$$N \hat{j} - F \hat{j} - mg \hat{j} = 0$$

$$N \hat{j} - (F + mg) \hat{j} = 0$$

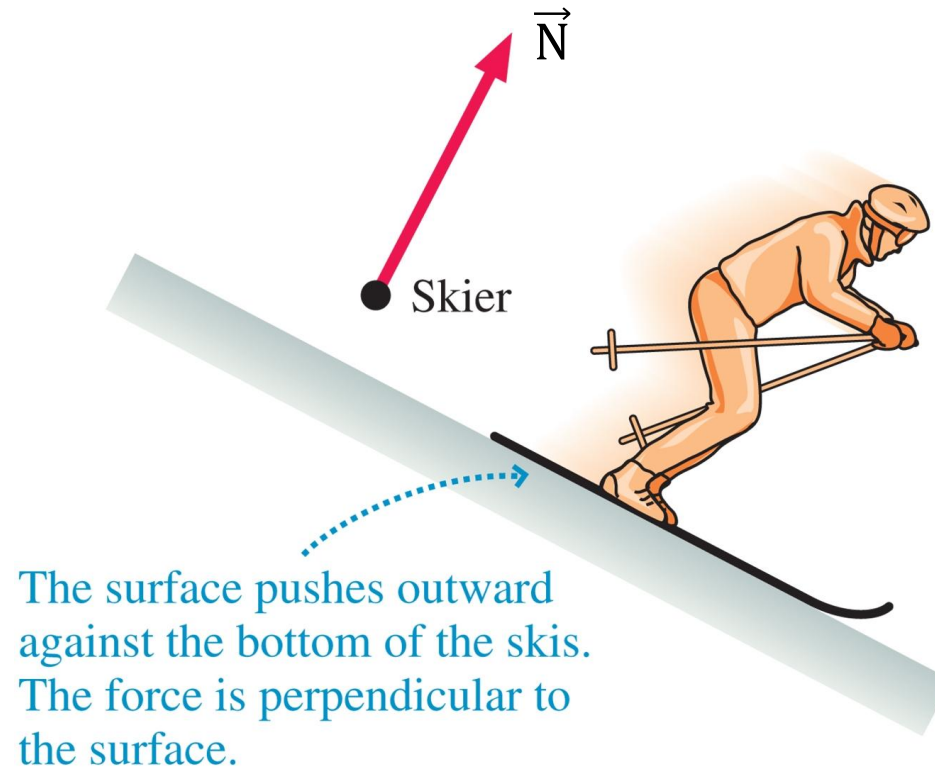
$$N = (F + mg)$$

$$\mathbf{N} \neq \mathbf{mg}$$



Normal Force (2)

- The **normal force** when the surface is inclined at an angle with the horizontal.



Tension Force ($\vec{T}$)

Definition: a force that develops in a rope, thread, or cable as it is stretched

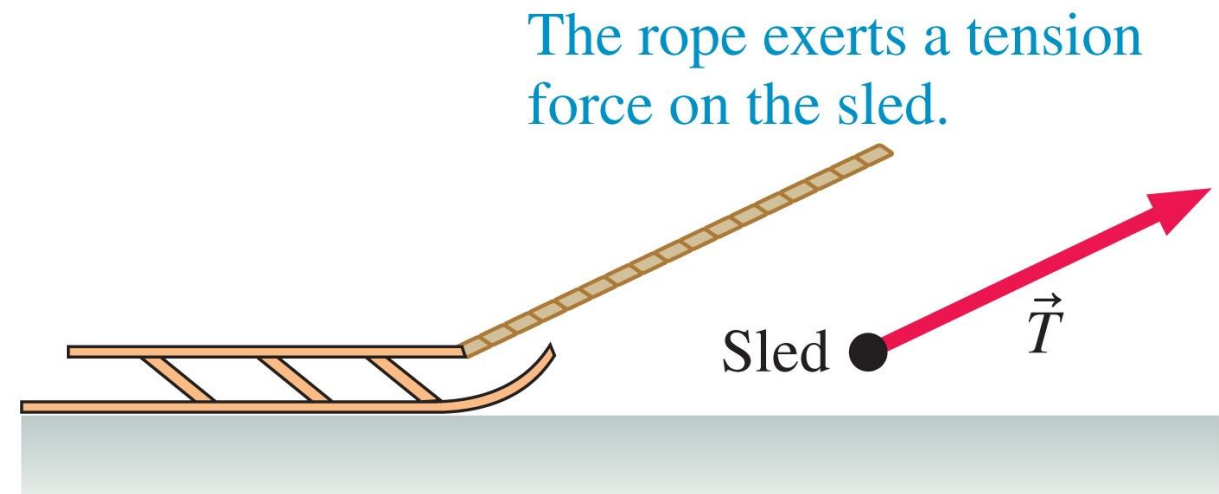
Variable: $\vec{T}$

Unit: Newtons (N)

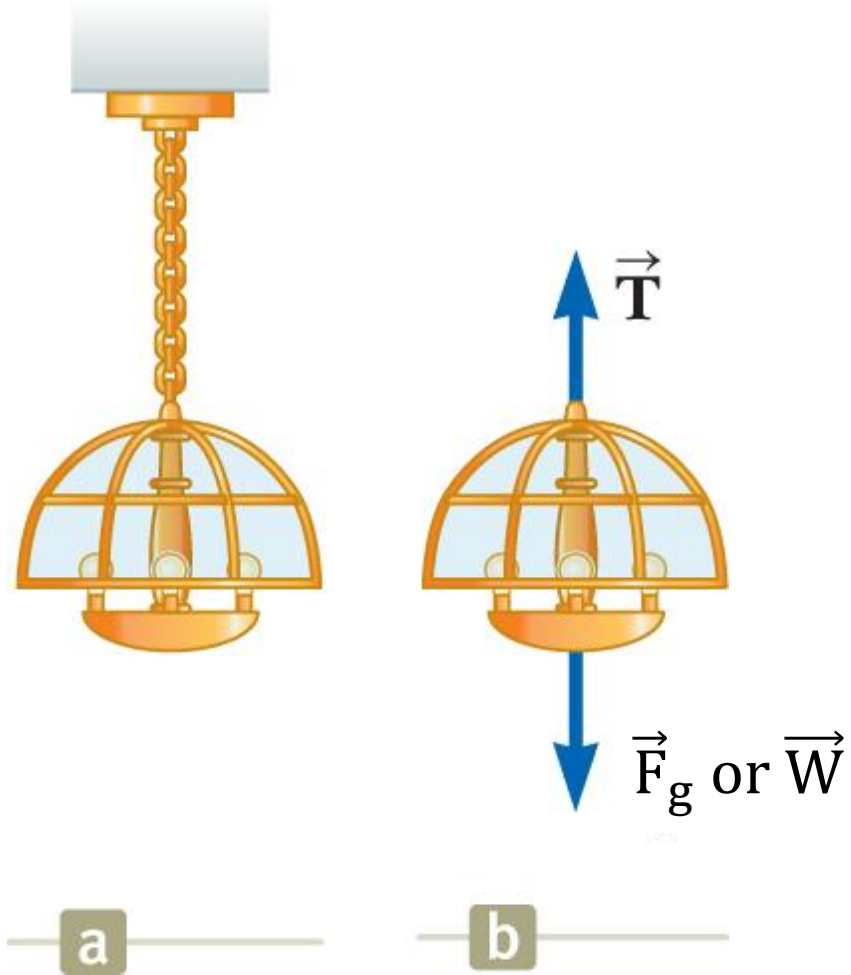
Equation: No specific equation

- Either is given or calculated from Newton's 2nd Law

Direction: Pointed away from force pulling on string



Tension Force



- If $\mathbf{a} = 0$: **particle in equilibrium**
model $\rightarrow$ net force on object = 0

$$\sum \vec{F} = m\vec{a}_y = 0$$

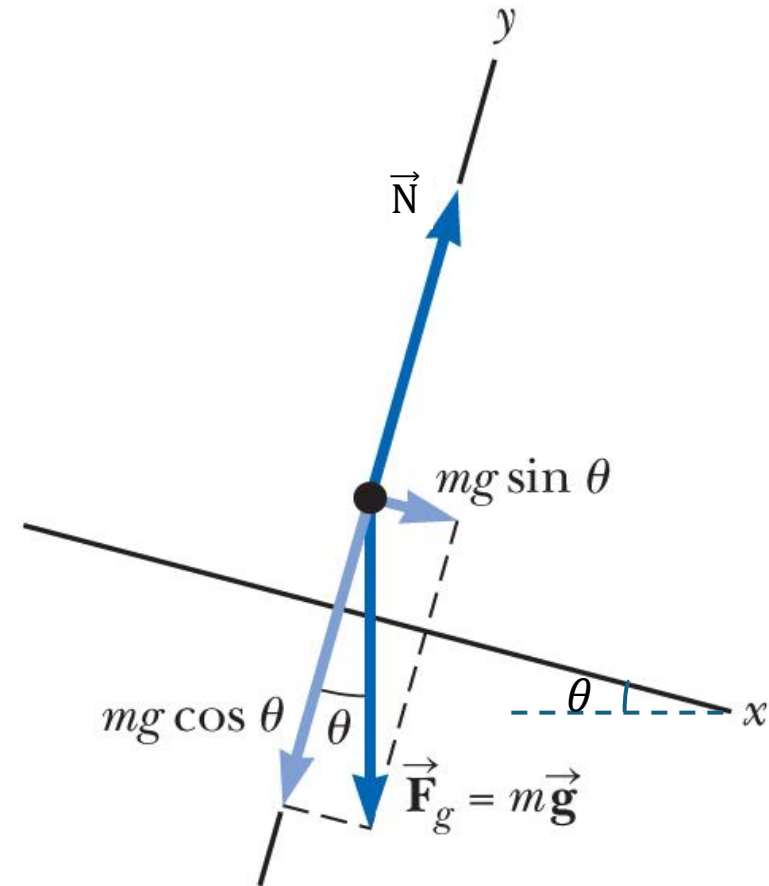
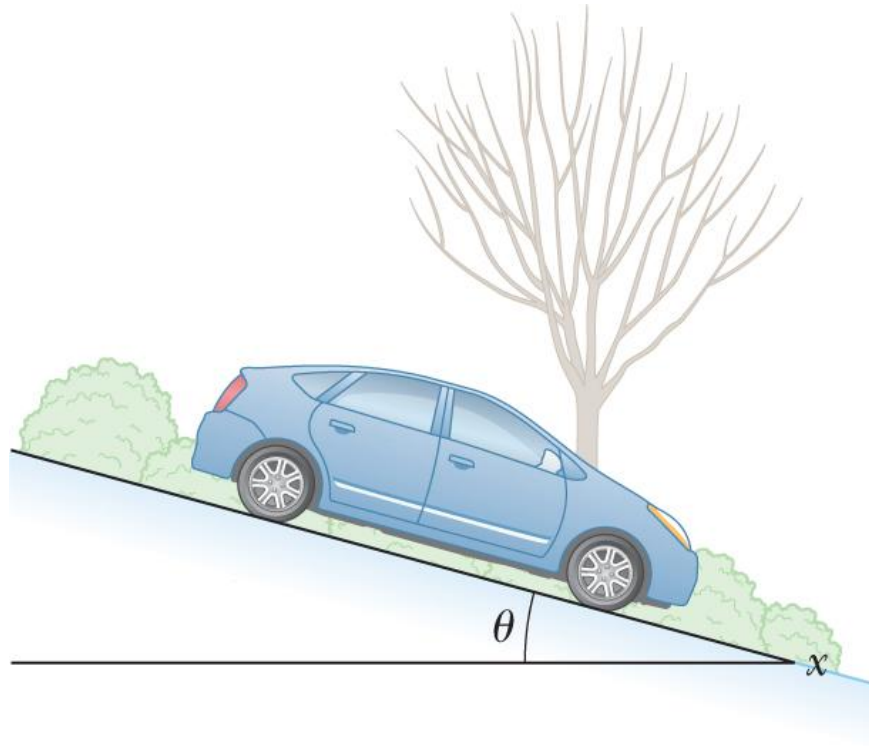
$$\sum F_y = 0$$

$$T - F_g(\text{or } W) = 0$$

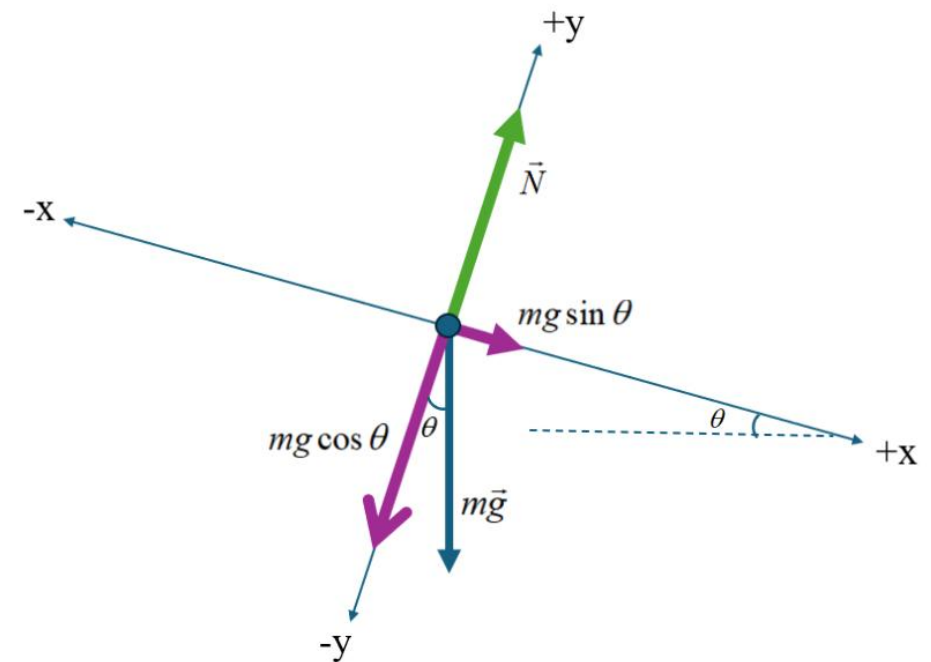
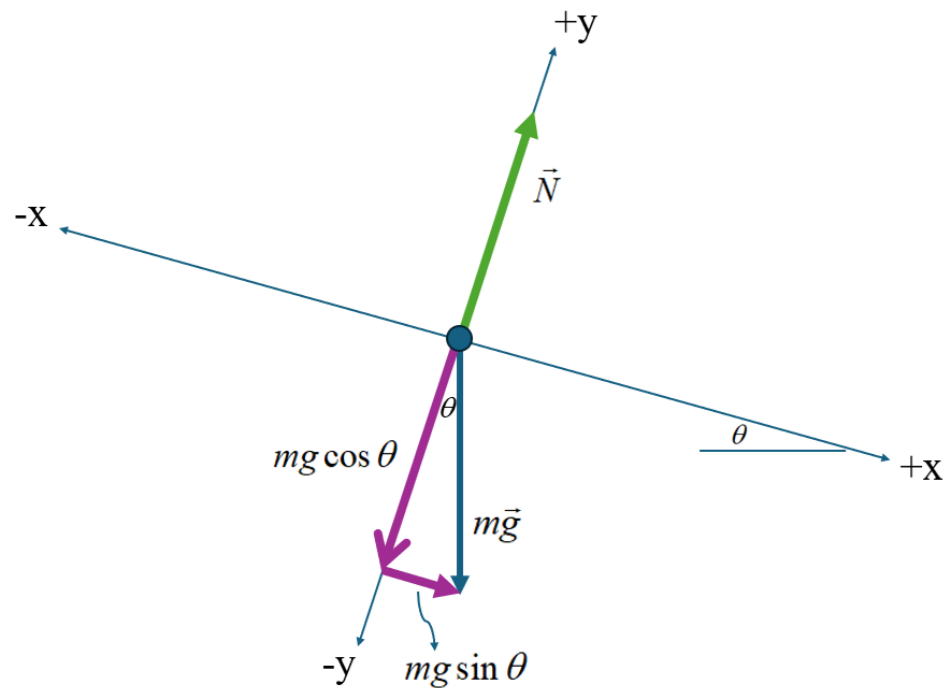
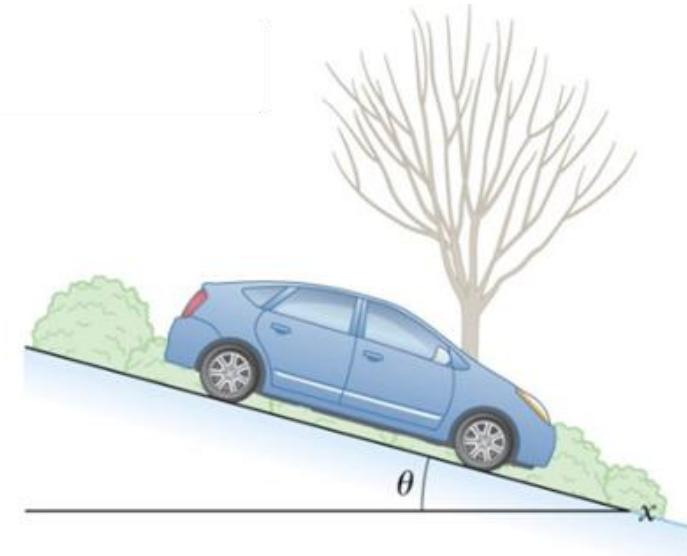
$$\text{or } T = F_g (\text{or } W)$$

Free Body Diagram (FBD)

Inclined plane

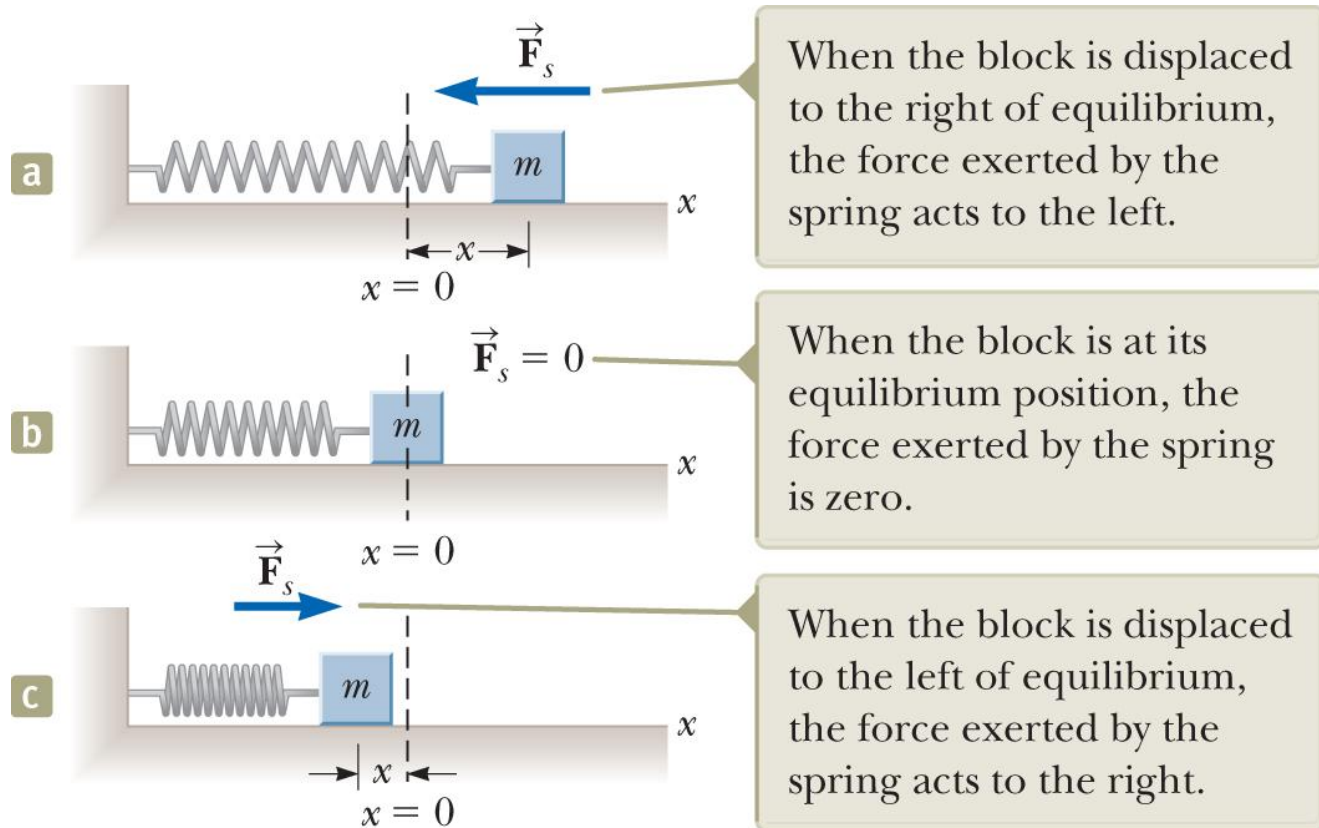


- The **normal force** acts perpendicular to the plane.
- The **gravitational force** acts straight down.
- Choose the coordinate system with **x along the incline** and y perpendicular to the incline.
- Replace the **force of gravity** with its **components**.



Spring Force ($\vec{F}_s$)

- Springs come in many forms. When deflected, they push or pull with a spring force.



Hooke's law:

$$F_s = -kx$$

$$F = -k(x - x_0)$$

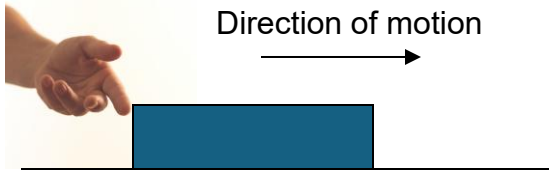
F_s = spring force

k = spring constant

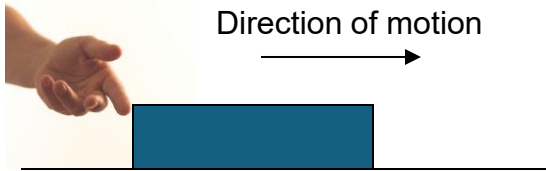
x = spring stretch or compression

Restoring force → always directed toward equilibrium position.

Free Body Diagram (FBD)

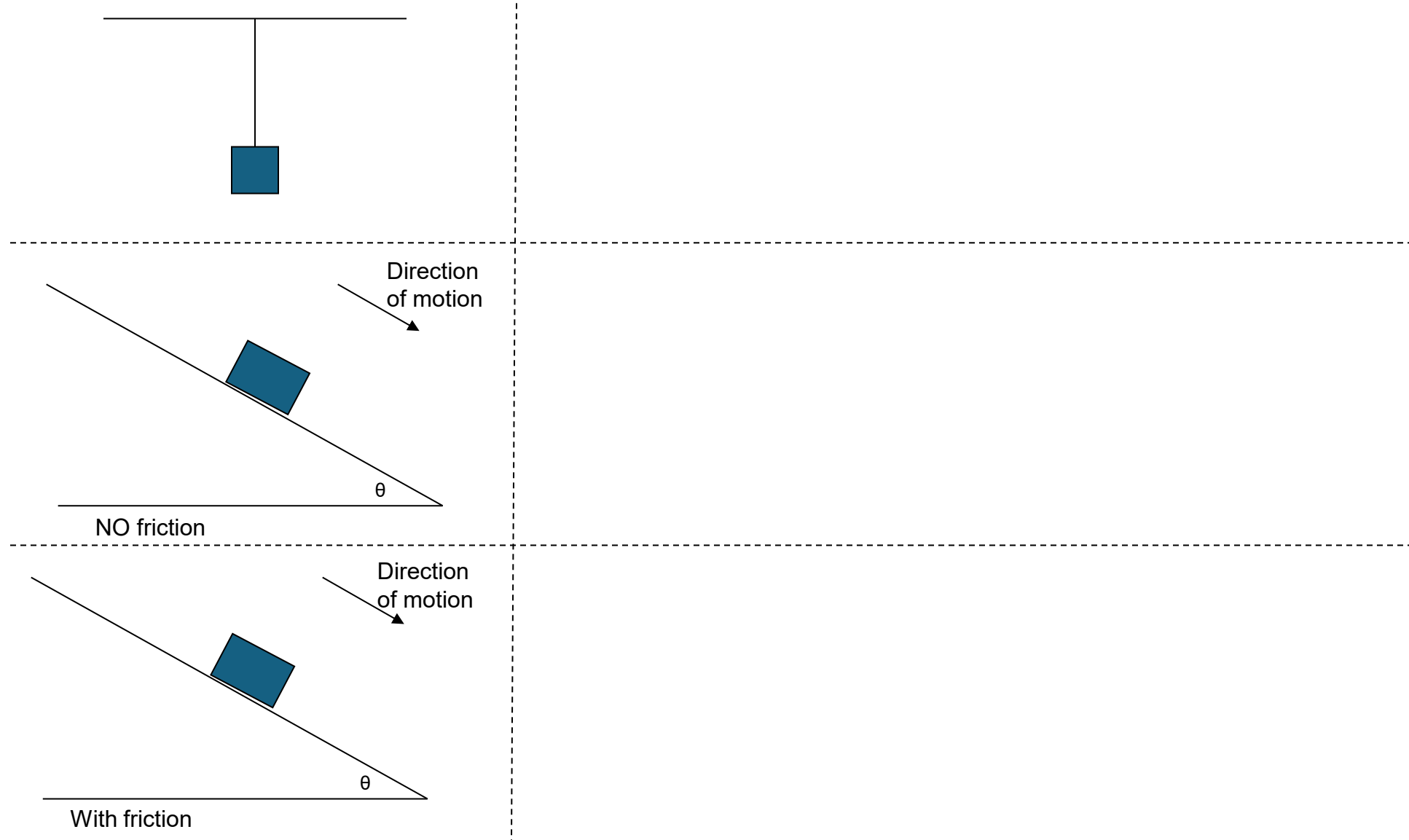


no friction



with friction

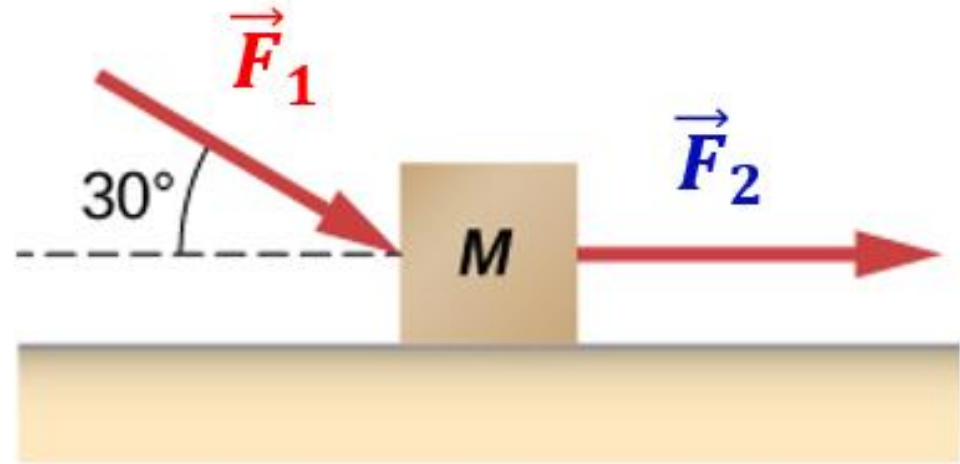
Free Body Diagram (FBD)



Problem solving (Newton's Law of Motion)

40) As shown in the figure, the block of mass 30.0 kg slides on the frictionless horizontal surface. If the two forces acting on the block are 30.0 N, what is the magnitude of the resulting **acceleration** of the block?

Answer: 1.87 m/s²



Problem solving (Newton's Law of Motion)

Given: mass (M) = 30.0 kg

$\vec{F} = 30.0 \text{ N}$ and $\theta = 30^\circ$

$a_x = ?$

By Newton's 2nd Law,

$$\Sigma F_x = m a_x$$

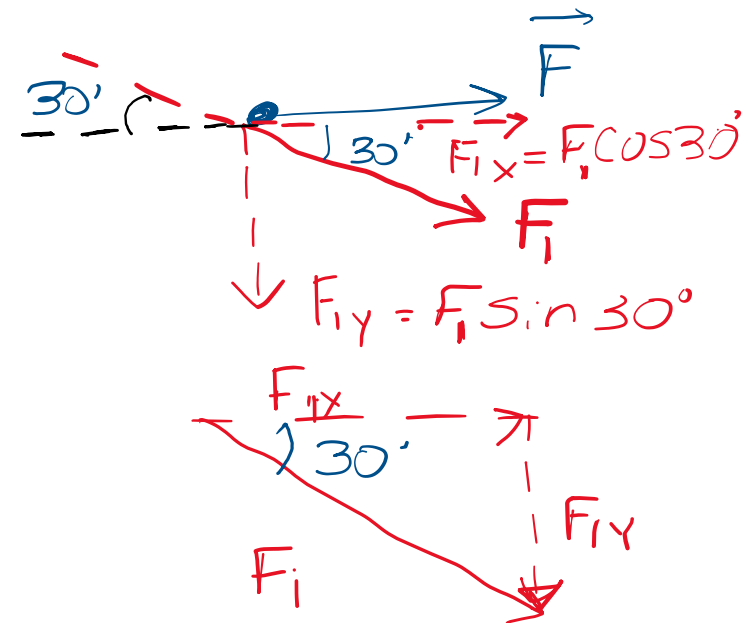
$$\vec{F}_2 + F_1 \cos 30^\circ = m a_x$$

$$\frac{F + F \cos 30^\circ}{m} = a_x$$

$$a_x = \frac{(30.0 \text{ N}) + (30.0 \text{ N}) \cos 30^\circ}{(30.0 \text{ kg})}$$

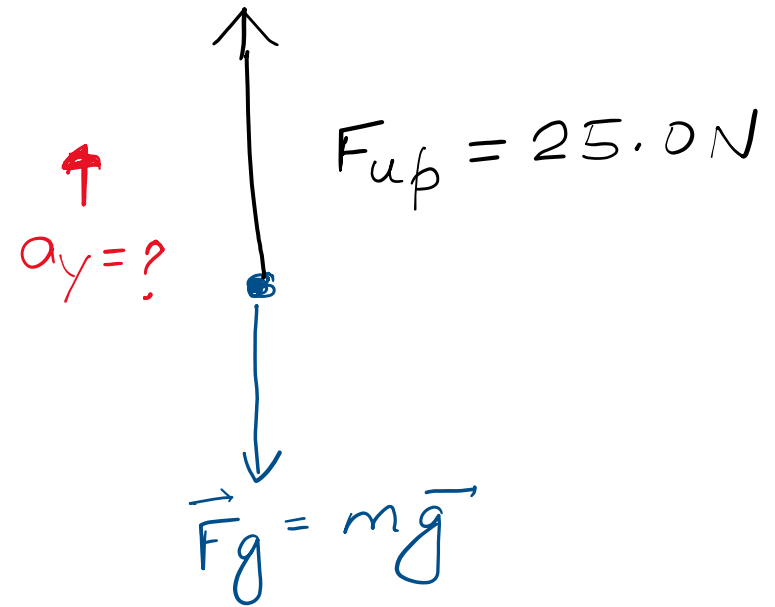
$$a_x = 1.97 \text{ m/s}^2$$

$$|\vec{F}_1| = |\vec{F}_2| = F$$



Problem solving (Newton's Law of Motion)

A body of mass 2.00 kg is pushed straight upward by a 25.0 N vertical force. What is its acceleration? (Ignore the friction)



Problem solving (Newton's Law of Motion)

Given: mass (m) = 2.00 kg

$$F_{up} = 25.0 \text{ N}$$

$$\begin{aligned}\vec{F}_g &= m\vec{g} \\ &= (2.00 \text{ kg})(9.80 \text{ m/s}^2) \\ &= 19.6 \text{ N}\end{aligned}$$

Using Newton's 2nd Law,

$$\Sigma F_y = ma_y$$

$$\vec{F}_{up} - \vec{F}_g = ma_y$$

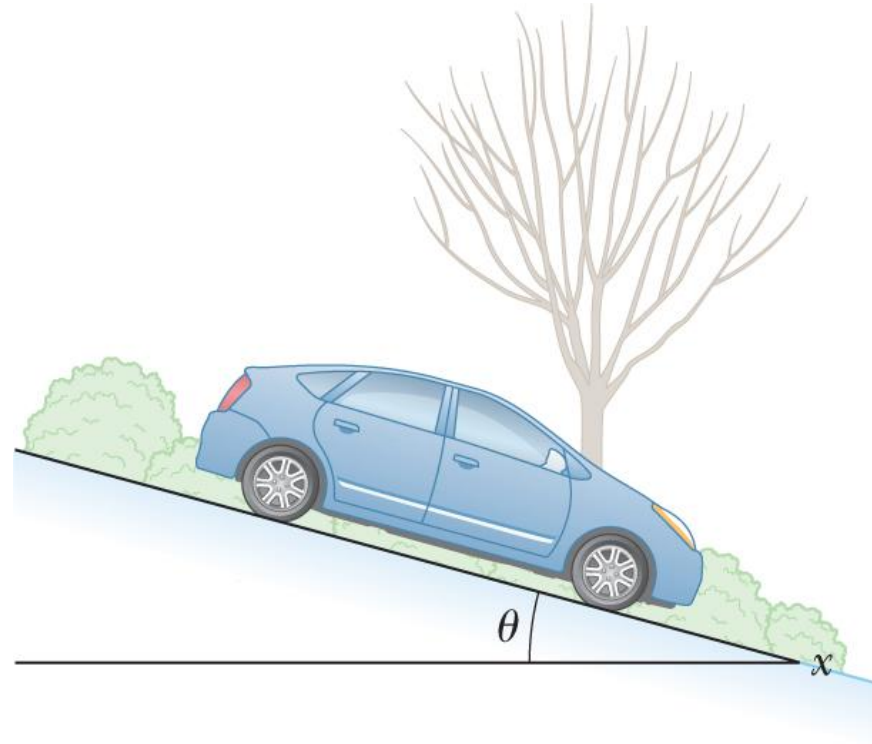
$$\frac{\vec{F}_{up} - \vec{F}_g}{m} = a_y$$

$$a_y = \frac{25.0 \text{ N} - 19.6 \text{ N}}{2.00 \text{ kg}} = 2.70 \text{ m/s}^2$$

Example: The Runway

A car of mass m is on an icy driveway inclined at an angle θ as in the figure.

(A) Find the acceleration of the car, assuming the driveway is frictionless.



Example: The Runway

From Newton's 2nd Law along x-axis,

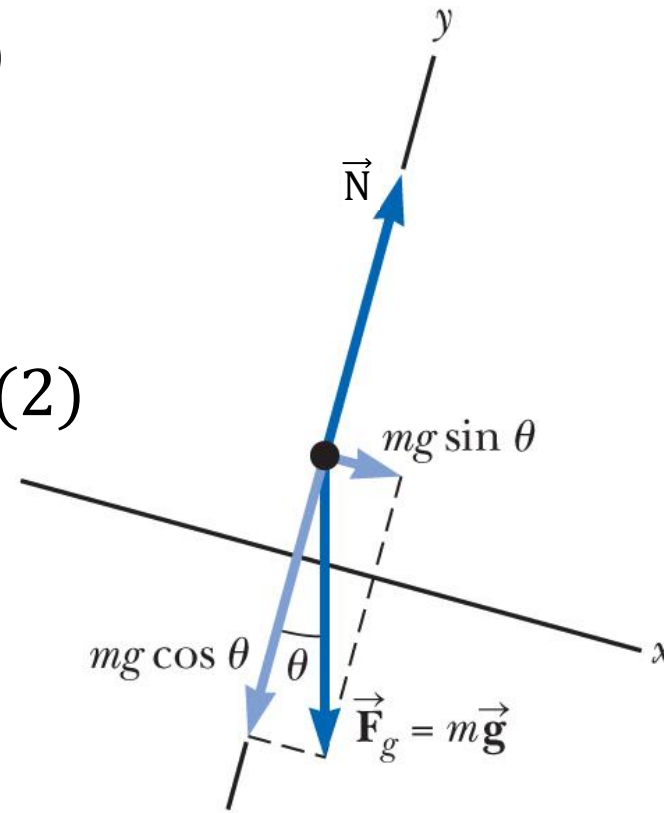
$$\sum F_x = mg \sin \theta = ma_x \dots (1)$$

From Newton's 2nd Law along y-axis,

$$\sum F_y = N - mg \cos \theta = ma_y \dots (2)$$

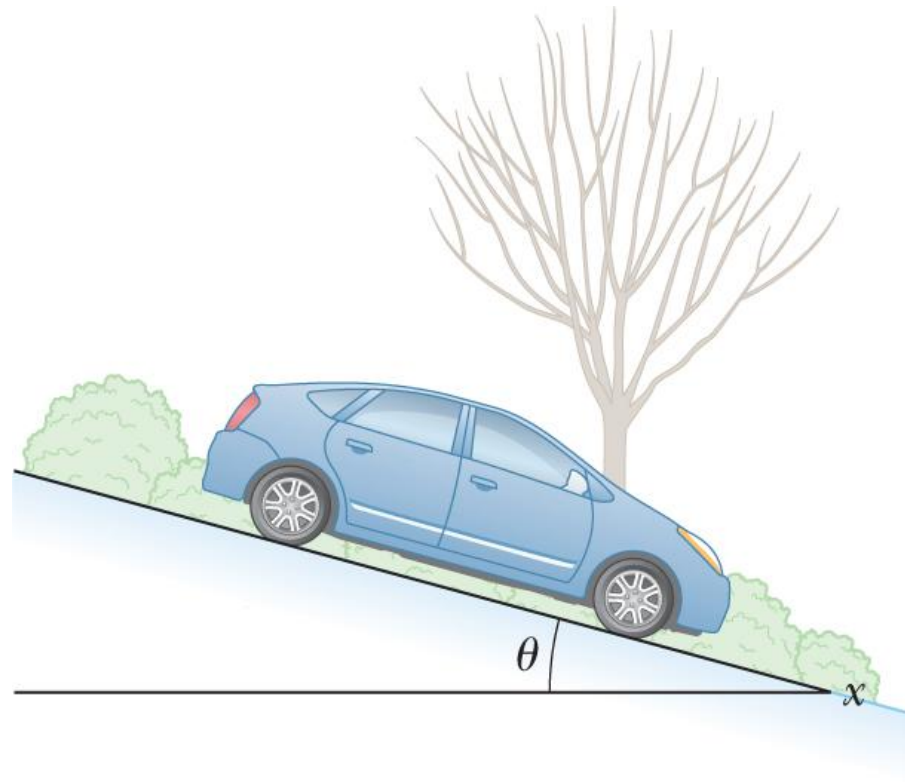
From equation (1)

$$a_x = g \sin \theta \dots (3)$$



Example: The Runway

(B) Suppose the car is released from rest at the top of the incline and the distance from the car's front bumper to the bottom of the incline is d . How **long** does it take the front bumper to reach the bottom of the hill, and what is the car's speed as it arrives there?



Example: The Runway

$$\mathbf{B} \text{ (i) } x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \rightarrow d = \frac{1}{2}a_x t^2$$

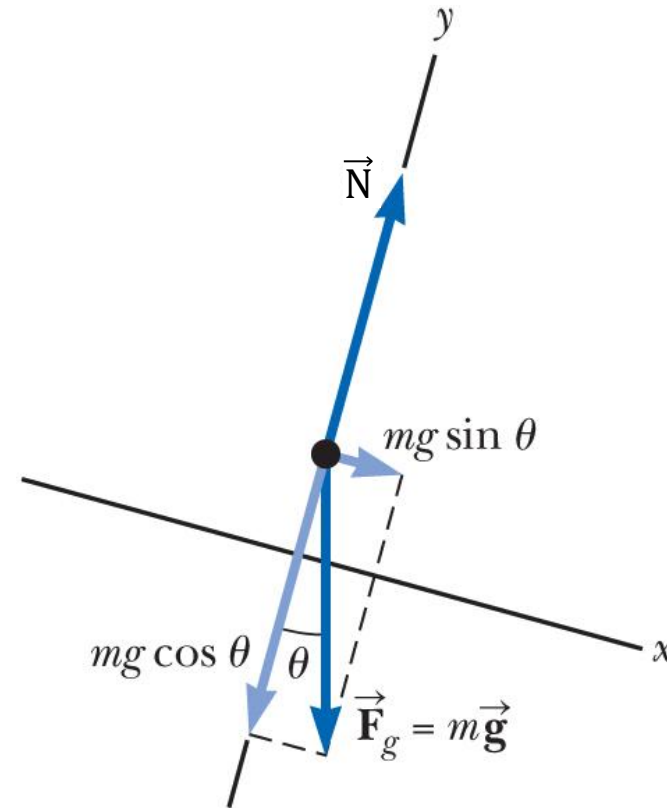
$$t = \sqrt{\frac{2d}{a_x}} = \boxed{\sqrt{\frac{2d}{g \sin \theta}}}$$

From equation (1)

$$\mathbf{B} \text{ (ii) } v_x^2 = v_{0x}^2 + 2a_x d$$

$$v_x^2 = 2a_x d$$

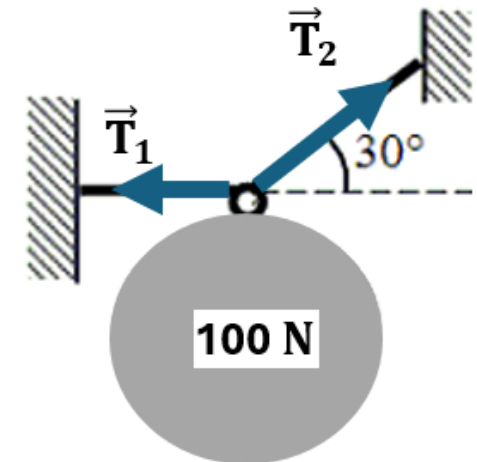
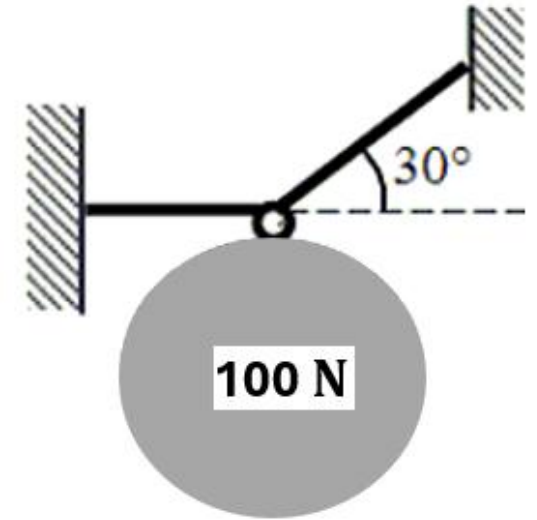
$$v_x = \sqrt{2a_x d} = \boxed{\sqrt{2gd \sin \theta}} \quad \text{From equation (1)}$$



Example 3a: Mass at Rest

A system of two cables supports a 100 N ball as shown. What is the tension on the right cable?

Answer = 200 N



Example 3a: Mass at Rest

Forces	x-component	y-component
T_1	$-T_1$	0
T_2	$T_2 \cos 30^\circ$	$T_2 \sin 30^\circ$
F_g	0	-100 N

$$\Sigma F_x = ma_x$$

$$T_1 + T_2 \cos 30^\circ = 0$$

$$T_1 = T_2 \cos 30^\circ \quad \text{--- (1)}$$

From (1) and (2)

$$T_1 = (200 \text{ N}) \cos 30^\circ$$

$$\boxed{T_1 = 173 \text{ N}}$$

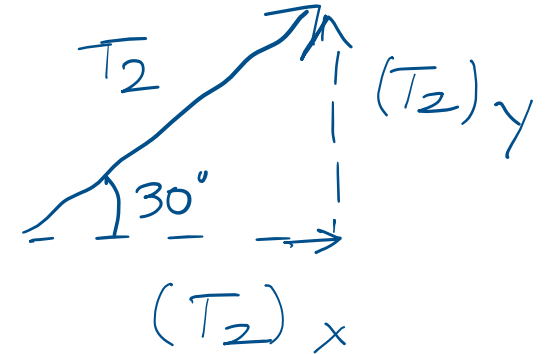
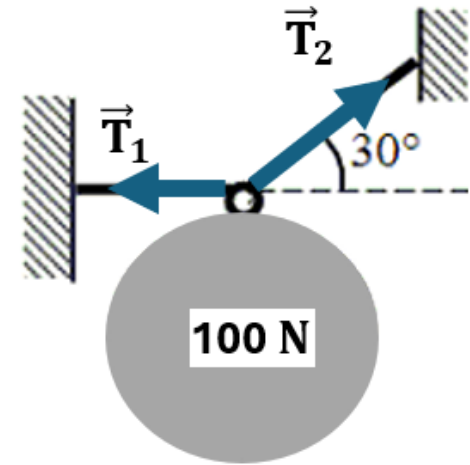
$$\Sigma F_y = ma_y$$

$$T_2 \sin 30^\circ - 100 \text{ N} = 0$$

$$T_2 \sin 30^\circ = 100 \text{ N}$$

$$T_2 = \frac{100 \text{ N}}{\sin 30^\circ}$$

$$\boxed{T_2 = 200 \text{ N}} \quad \text{--- (2)}$$



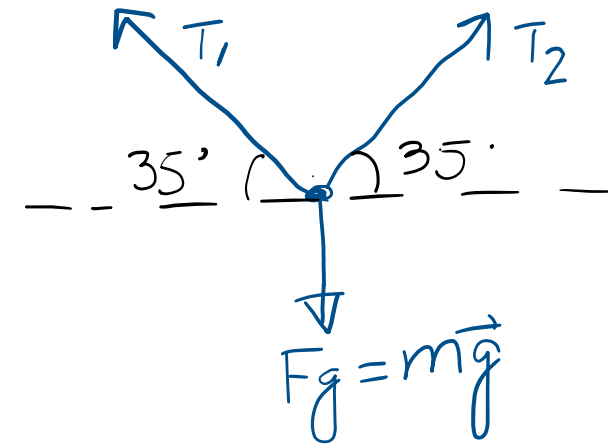
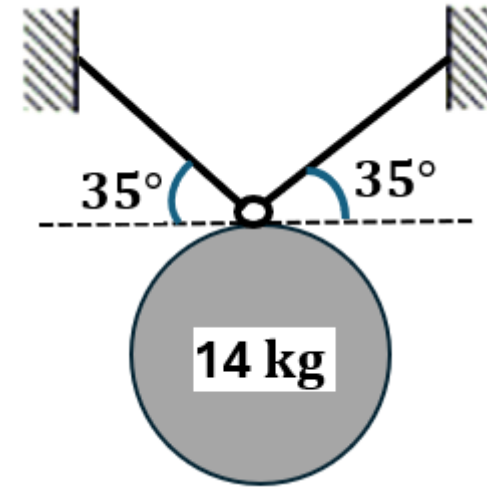
$$(T_2)_x = T_2 \cos 30^\circ$$

$$(T_2)_y = T_2 \sin 30^\circ$$

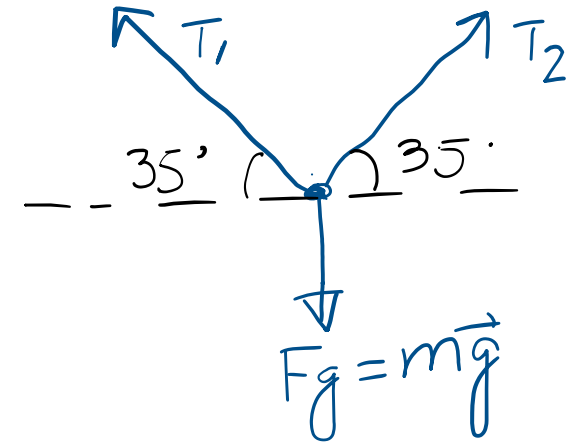
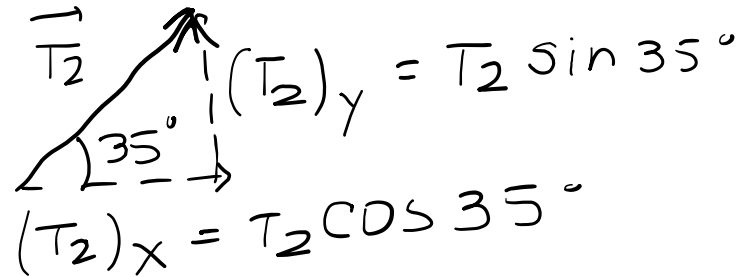
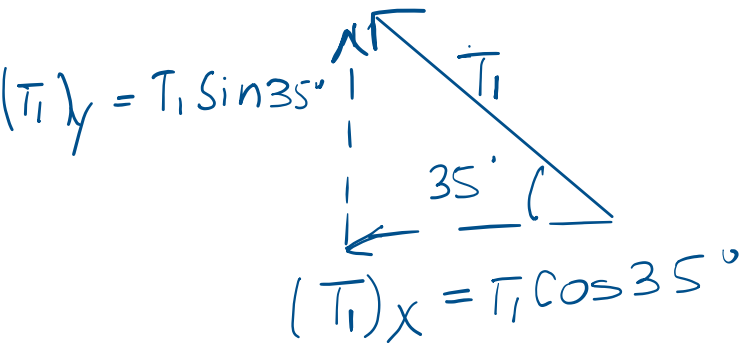
Example 3a: Mass at Rest

A system of two cables supports a 14.0 kg ball as shown. What is the tension on the right cable?

Answer = 120 N



Example 3a: Mass at Rest



Force	x-comp	y-comp
T_1	$-T_1 \cos 35^\circ$	$T_1 \sin 35^\circ$
T_2	$T_2 \cos 35^\circ$	$T_2 \sin 35^\circ$
F_g	0	$-mg$

$$\Sigma F_x = m a_x$$

$$-T_1 \cos 35^\circ + T_2 \cos 35^\circ + 0 = 0$$

$$\boxed{T_1 = T_2} \quad \text{--- (1)}$$

$$\Sigma F_y = m a_y$$

$$T_1 \sin 35^\circ + T_2 \sin 35^\circ - mg = 0$$

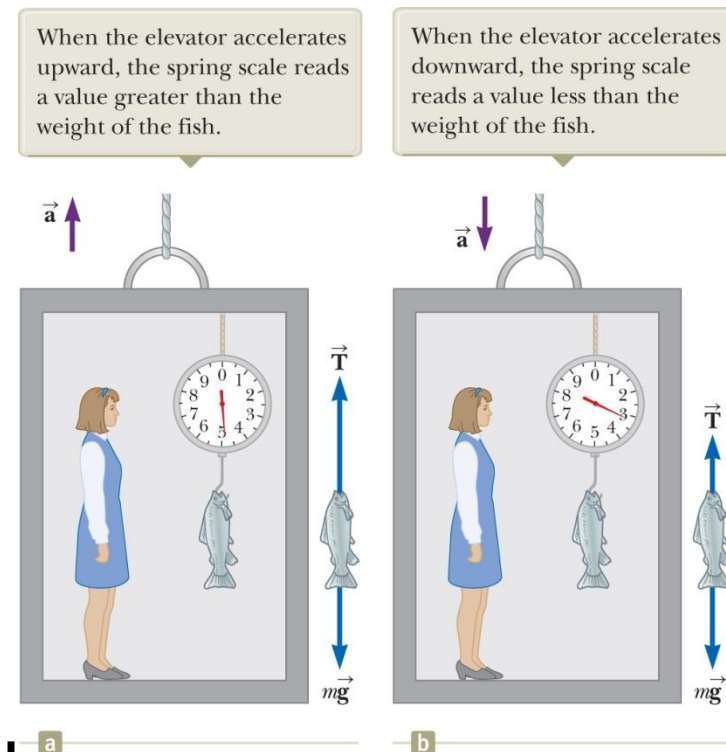
$$2 T_1 \sin 35^\circ = mg$$

$$\boxed{T_1 = ?}$$

Example: Weighing a Fish in an Elevator

A person weighs a fish of mass m on a spring scale attached to the ceiling of an elevator as illustrated in the figure.

(A) Show that if the elevator accelerates either upward or downward, the spring scale gives a reading that is different from the weight of the fish.



Example 5.7: Weighing a Fish in an Elevator

Example: Weighing a Fish in an Elevator

Case 1 (If elevator at rest or moving with constant v)

$$\sum F_y = 0$$

$$T - F_g = 0$$

$$\Rightarrow T = F_g = mg$$

Case 2

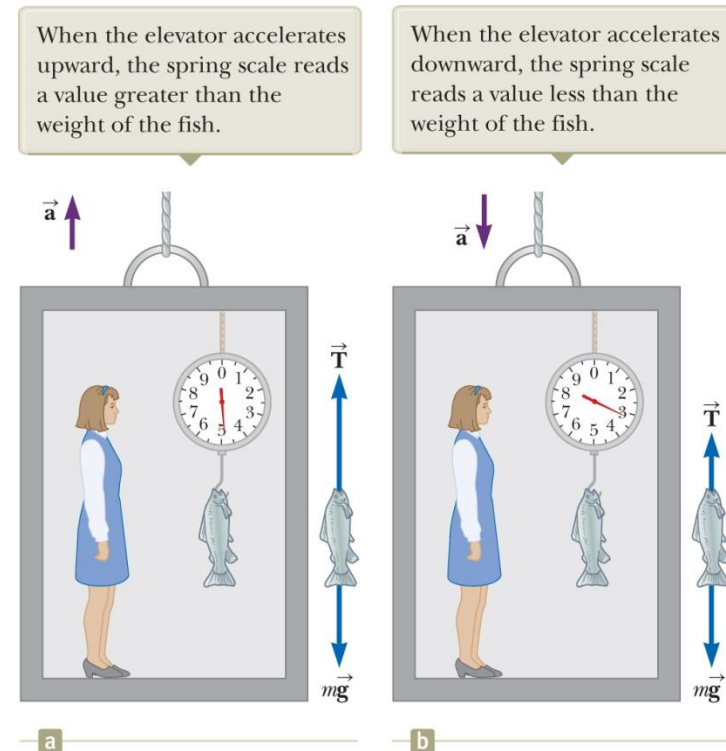
(If elevator is *accelerating up or down*)

$$\sum F_y = ma_y$$

$$T - mg = ma_y$$

$$T = mg + ma_y \quad \text{accelerating upward } a_y = \text{positive}$$

$$T = mg - ma_y \quad \text{accelerating downward } a_y = \text{negative}$$



Example: Weighing a Fish in an Elevator

(B) Evaluate the scale readings for a 40.0 N fish if the elevator moves with an acceleration $a_y = \pm 2.00 \text{ m/s}^2$.

$$F_g = mg = 40.0 \text{ N}$$

$$\Rightarrow m = \frac{40.0 \text{ N}}{g} = \frac{40.0 \text{ N}}{9.80} = 4.08 \text{ kg}$$

accelerating upward $a_y = \text{positive}$

$$T = mg + ma_y$$

$$= 40.0 + 4.08 (2.00)$$

$$T = \boxed{48.2 \text{ N}}$$

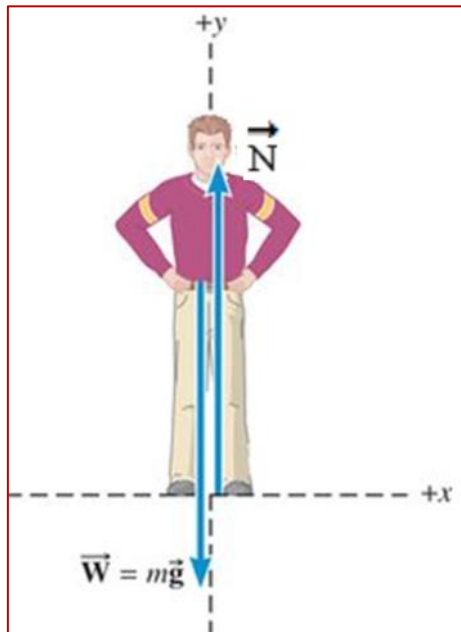
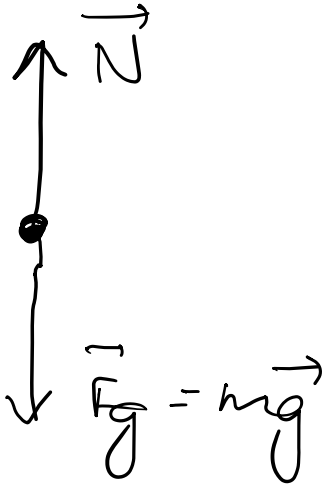
accelerating downward $a_y = \text{negative}$

$$T = mg - ma_y$$

$$= 40.0 - 4.08 (2.00)$$

$$T = \boxed{31.8 \text{ N}}$$

Example: Apparent weight



If the elevator is accelerating upward,

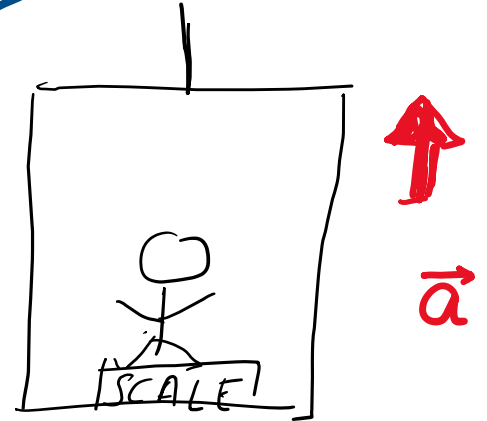
$$\Sigma F_y = ma_y$$

$$\vec{N} - \vec{F}_g = ma$$

$$\vec{N} = \vec{F}_g + ma$$

Reading on the scale (OR)

Your apparent weight appears more than your actual weight.

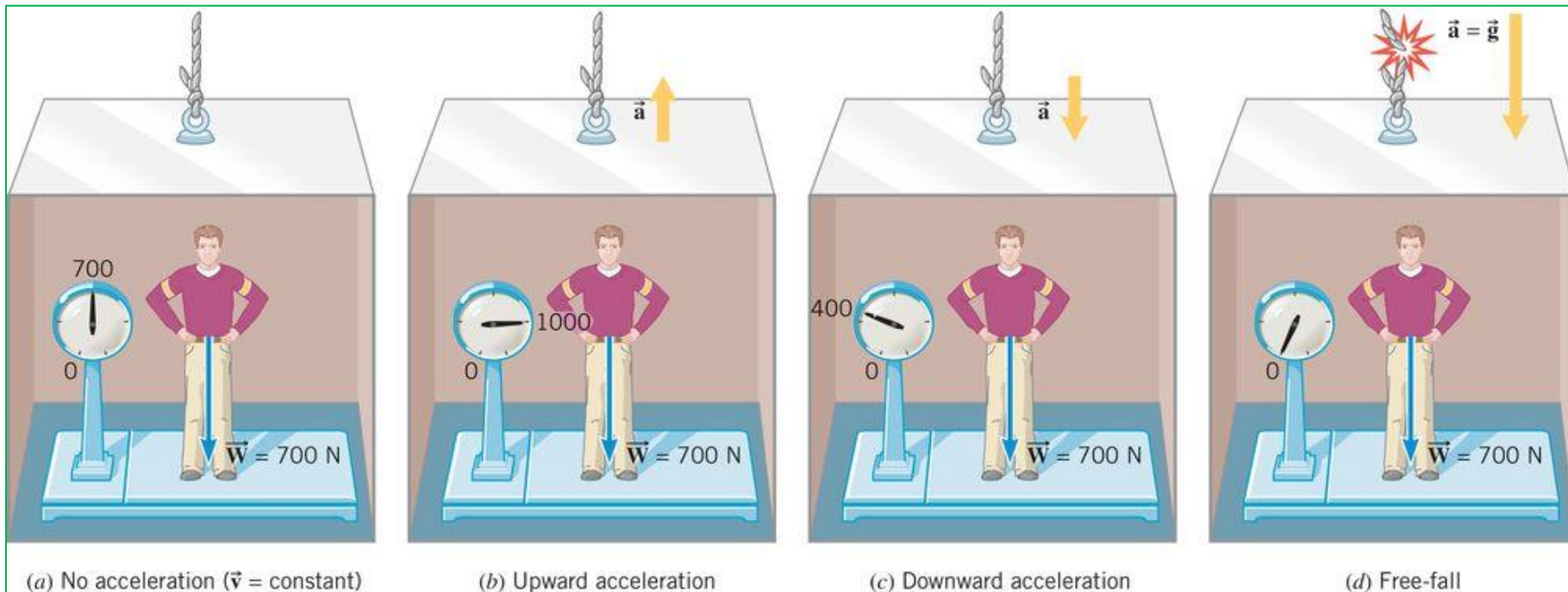
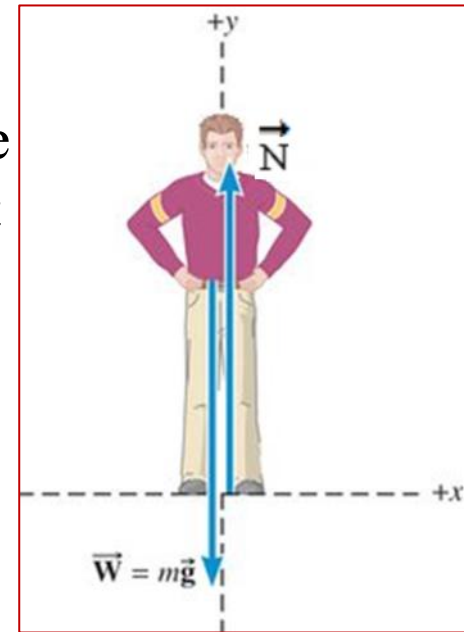


APPARENT WEIGHT

Consider the scale in the elevator in the figure below. The apparent weight of an object is the reading of the scale. It is equal to the normal force the man exerts on the scale.

$$\sum \vec{F}_y = +N - mg = m\vec{a}_y$$

To determine the **apparent weight**, solve for the normal force.



Summary: Newton's Law

■ Newton's 1st & 2nd Laws

- Draw and interpret a free-body diagram.
- Describe the consequences of no force or balanced forces on an object.
- Write Newton's 2nd Law for a physical scenario.
- Use Newton's 2nd Law to solve a problem
 - With balanced forces.
 - With unbalanced forces.
- Use Newton's Laws to predict the motion of an object.

■ Other Forces and Newton's 3rd Laws

- Use Newton's 3rd Law to Identify force pairs.
- Solve a problems involving static friction, kinetic friction, gravity, tension, and normal force.